

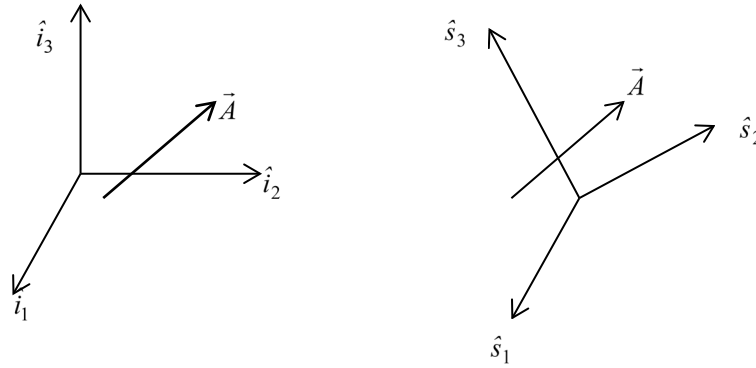
Spacecraft Attitude Control and Determination

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1. Kinematics and Dynamics of Angular Motion

Coordinate and Rotations



$$\vec{A}^i = A_{i1}\hat{i}_1 + A_{i2}\hat{i}_2 + A_{i3}\hat{i}_3$$

$$\vec{A}^s = A_{s1}\hat{s}_1 + A_{s2}\hat{s}_2 + A_{s3}\hat{s}_3$$

$\vec{A}$ vector is usually represented by its coordinates with respect to a given frame of reference.

So, a vector $\vec{A}$ has an infinity of representations in terms of the basis vectors of different reference frame. The coordinates A_i and A_s are different, but they represent the same vector.

To calculate the components of a vector $\vec{A}$

$$\begin{aligned} A_{i1} &= \vec{A} \cdot \hat{i}_1 \\ &= A_{s1}\hat{s}_1 \cdot \hat{i}_1 + A_{s2}\hat{s}_2 \cdot \hat{i}_1 + A_{s3}\hat{s}_3 \cdot \hat{i}_1 \end{aligned}$$

$$\begin{aligned} A_{i2} &= \vec{A} \cdot \hat{i}_2 \\ &= A_{s1}\hat{s}_1 \cdot \hat{i}_2 + A_{s2}\hat{s}_2 \cdot \hat{i}_2 + A_{s3}\hat{s}_3 \cdot \hat{i}_2 \end{aligned}$$

$$\begin{aligned} A_{i3} &= \vec{A} \cdot \hat{i}_3 \\ &= A_{s1}\hat{s}_1 \cdot \hat{i}_3 + A_{s2}\hat{s}_2 \cdot \hat{i}_3 + A_{s3}\hat{s}_3 \cdot \hat{i}_3 \end{aligned}$$

$$\rightarrow \begin{bmatrix} A_{i1} \\ A_{i2} \\ A_{i3} \end{bmatrix} = \begin{bmatrix} \hat{i}_1 \cdot \hat{s}_1 & \hat{i}_1 \cdot \hat{s}_2 & \hat{i}_1 \cdot \hat{s}_3 \\ \hat{i}_2 \cdot \hat{s}_1 & \hat{i}_2 \cdot \hat{s}_2 & \hat{i}_2 \cdot \hat{s}_3 \\ \hat{i}_3 \cdot \hat{s}_1 & \hat{i}_3 \cdot \hat{s}_2 & \hat{i}_3 \cdot \hat{s}_3 \end{bmatrix} \begin{bmatrix} A_{s1} \\ A_{s2} \\ A_{s3} \end{bmatrix}$$

$\uparrow R^{si}$,
“Direction Cosine Matrix” or “Rotational Matrix”

A rotational matrix is simply the component of one set of basis vectors with respect to the other set of basis vector.

$$\vec{A}^i = R^{si} \vec{A}^s \quad \text{from s to i frame}$$

$$\begin{aligned} \vec{A}^s &= (R^{si})^T \vec{A}^i \\ &= R^{is} \vec{A}^i \end{aligned}$$

* Rotation matrix is orthogonal matrix ($Q^T Q = I$)
 o)므로 $(R^{si})^{-1} = (R^{si})^T$

Velocity

$$\vec{A}^i = R^{si} \vec{A}^s$$

$$\text{inertial frame derivative : } \frac{d\vec{A}^i}{dt} = \frac{d}{dt}(R^{si} \vec{A}^s) = \dot{R}^{si} \vec{A}^s + R^{si} \dot{\vec{A}}^s$$

where $\dot{\vec{A}}^s$: body (s) frame derivative

Transform this result back to s-frame.

$$\begin{aligned} \frac{{}^i d\vec{A}^s}{dt} &= \left(\frac{{}^i d\vec{A}^s}{dt} \right)^s = R^{is} \frac{d\vec{A}^i}{dt} \\ &= R^{is} \left(\dot{R}^{si} \vec{A}^s + R^{si} \dot{\vec{A}}^s \right) = R^{is} \dot{R}^{si} \vec{A}^s + R^{is} R^{si} \dot{\vec{A}}^s \end{aligned}$$

$$R^{is} \cdot R^{si} = I \rightarrow$$

$$(R^{si})^T \cdot R^{si} = I$$

$$\rightarrow \frac{d}{dt} \left[(R^{si})^T \cdot R^{si} \right] = \frac{dI}{dt} = 0$$

$$\text{let } R^{si} = R$$

$$\dot{R}^T R + R^T \dot{R} = 0$$

$$R^T \dot{R} = -\dot{R}^T R$$

$$= -[R^T \dot{R}]^T ; \text{ skew symmetric matrix}$$

$$R^{siT} \dot{R}^{si} = R^{is} \dot{R}^{si} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$$

$$R^{is} \dot{R}^{si} \vec{A}^s = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \begin{bmatrix} A_{s1} \\ A_{s2} \\ A_{s3} \end{bmatrix}$$

$$= \begin{bmatrix} \omega_2 A_{s3} - \omega_3 A_{s2} \\ \omega_3 A_{s1} - \omega_1 A_{s3} \\ \omega_1 A_{s2} - \omega_2 A_{s1} \end{bmatrix}$$

$$= \vec{\omega}^{si} \times \vec{A}^s$$

$$\text{where } \vec{\omega}^{si} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \text{ angular velocity of s-frame with respect to i-}$$

frame (expressed in s-frame)

$$\frac{{}^i d \vec{A}^s}{dt} = \dot{\vec{A}}^s + \vec{\omega}^{si} \times \vec{A}^s$$

In general,

$$\frac{{}^i d}{dt} () = \frac{{}^s d}{dt} () + \vec{\omega}^{si} \times ()$$

Spacecraft Orientation

Recall $\vec{1} = R^{si} \vec{s}$

$$R^{is} \dot{R}^{si} = R^{is} (\dot{R}^{is})^T = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \equiv [\tilde{\omega}]$$

$$(\dot{R}^{is})^T = (R^{is})^T \tilde{\omega}$$

$$\dot{R}^{is} = \left[(R^{is})^T \tilde{\omega} \right]^T = \tilde{\omega}^T \left[(R^{is})^T \right]^T = -\tilde{\omega} R^{is}$$

$$\therefore \dot{R}^{is} = -\tilde{\omega} R^{is}$$

This differential equation describing the change the orientation of the spacecraft due to $\vec{\omega}$ (angular velocity).

Initial conditions have to be specified.

If i-frame and s-frame are coincident at t=0

$$R^{is}(0) = I$$

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The 9 scalar equations exist. Only 3 are independent.

The other 6 are related through orthogonality constraints.

Parameterization of R

- i) Direction cosin Classical Euler angles
- ii) Euler angles : Roll-pitch-yaw Euler angles
- iii) Euler axis (Eigen axis) and principal angle
- iv) Euler parameter (Quaternions) ; to remove singularity
- v) other parameters

Classical Euler angle

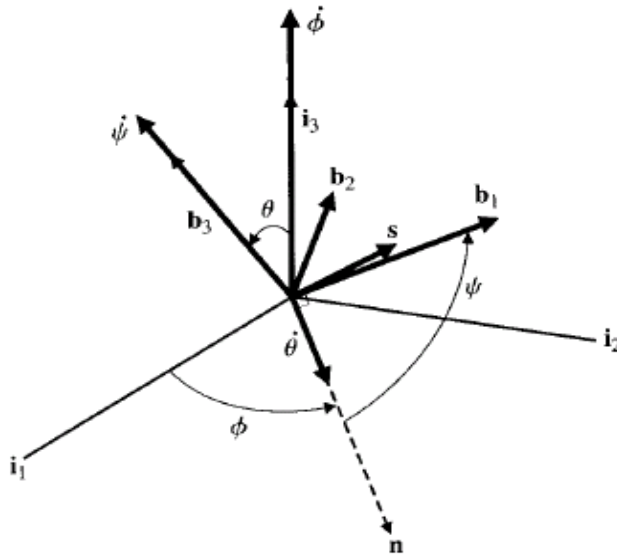
Classical set : ϕ, θ, ψ

ϕ : right ascension of the ascending node

θ : orbit inclination

ψ : argument of periapsis

(i_1, i_2, i_3) inertial frame, (b_1, b_2, b_3) body frame



$R^{ib} = R_3(\psi)R_1(\theta)R_3(\phi)$: rotation matrix from i-frame to b-frame

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{i}_1 \\ \hat{i}_2 \\ \hat{i}_3 \end{bmatrix}$$

$$\text{Let } R^{ib} = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

$$= \begin{bmatrix} \cos \phi \cos \psi - \sin \phi \cos \theta \sin \psi & \sin \phi \cos \psi + \cos \phi \cos \theta \sin \psi & \sin \theta \sin \psi \\ -\cos \phi \sin \psi - \sin \phi \cos \theta \cos \psi & -\sin \phi \sin \psi + \cos \phi \cos \theta \cos \psi & \sin \theta \cos \psi \\ \sin \phi \sin \theta & -\cos \phi \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \hat{i}_1 \\ \hat{i}_2 \\ \hat{i}_3 \end{bmatrix}$$

$$\Rightarrow \theta = \cos^{-1}(C_{33}), \quad \phi = \tan^{-1}(C_{31}/-C_{32}), \quad \psi = \tan^{-1}(C_{13}/C_{23})$$

Angular velocity

$$\vec{\omega}^{\text{bi}} = \dot{\phi} \hat{i}_3 + \dot{\theta} \hat{n} + \dot{\psi} \hat{b}_3$$

$$\hat{n} = \cos \psi \hat{b}_1 - \sin \psi \hat{b}_2$$

$$\hat{i}_3 = \cos \theta \hat{b}_3 + \sin \theta \hat{s}, \quad \text{let } \hat{s} \text{ vector line at } 90^\circ \text{ to } \hat{n}$$

$$\hat{s} = \sin \psi \hat{b}_1 + \cos \psi \hat{b}_2$$

$$\Rightarrow \vec{\omega}^{\text{bi}} = (\dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi) \hat{b}_1 + (\dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi) \hat{b}_2 + (\dot{\psi} + \dot{\phi} \cos \theta) \hat{b}_3$$

$$\Leftrightarrow \vec{\omega}^{\text{bi}} = \begin{bmatrix} \omega_1 \hat{b}_1 \\ \omega_2 \hat{b}_2 \\ \omega_3 \hat{b}_3 \end{bmatrix}$$

$$\Rightarrow \omega_1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi$$

$$\omega_2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \quad \text{expressed in the body-frame}$$

$$\omega_3 = \dot{\psi} + \dot{\phi} \cos \theta$$

$$\therefore \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} \sin \theta \sin \psi & \cos \psi & 0 \\ \sin \theta \cos \psi & -\sin \psi & 0 \\ \cos \theta & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \frac{1}{\sin \theta} \begin{bmatrix} \sin \psi & \cos \psi & 0 \\ \sin \theta \cos \psi & -\sin \theta \sin \psi & 0 \\ -\cos \theta \sin \psi & -\cos \theta \cos \psi & \sin \theta \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

Singular when $\theta = 0$ or $\pm \pi$

They become three coupled differential equations for the Euler angles.

The second half of equations of motion of a rigid body.

Their left $(\omega_1, \omega_2, \omega_3)$ are obtained from the solution to Euler's equations.

Their solution will furnish the orientation of the body as a function of time.

TABLE: Euler Angle Geometric and Kinematic Formula Summary

Direction Cosine Parameterization: $[C(\theta_1, \theta_2, \theta_3)] = [M_\gamma(\theta_3)] [M_\beta(\theta_2)] [M_\alpha(\theta_1)]$

Angular velocity/Angular Rate Transform: $\{\omega\} = [R(\theta_2, \theta_3)] \{\dot{\theta}\}$

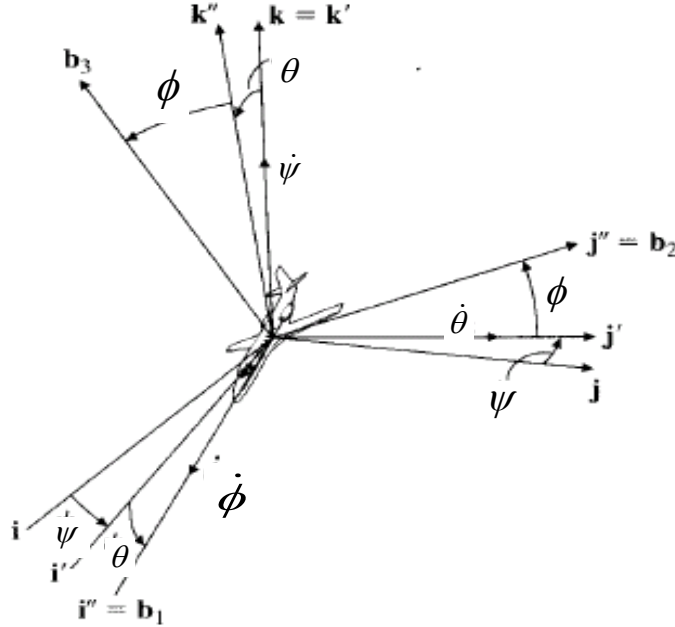
Abbreviations: $c_i = \cos(\theta_i)$, $s_i = \sin(\theta_i)$, $c^{-1}(\) = \arccos(\)$, $s^{-1}(\) = \arcsin(\)$, $t^{-1}(\) = \arctan(\)$

α - β - γ Rotation Sequence	R	R ⁻¹	Angles as Function of Direction Cosines	Singular at $\theta_2 =$
1-2-1	$\begin{bmatrix} c_2 & 0 & 1 \\ s_2 s_3 & c_3 & 0 \\ s_2 c_3 & -s_3 & 0 \end{bmatrix}$	$\frac{1}{s_2} \begin{bmatrix} 0 & s_3 & c_3 \\ 0 & s_2 c_3 & -s_2 c_3 \\ s_2 & -c_2 s_3 & -c_2 c_3 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{12} / -C_{13})$ $\theta_2 = c^{-1}(C_{11})$ $\theta_3 = t^{-1}(C_{21} / C_{31})$	$0, \pm\pi$
1-2-3	$\begin{bmatrix} c_2 c_3 & s_3 & 0 \\ -c_2 s_3 & c_3 & 0 \\ s_2 & 0 & 1 \end{bmatrix}$	$\frac{1}{c_2} \begin{bmatrix} c_3 & -s_3 & 0 \\ c_2 s_3 & c_2 c_3 & 0 \\ -s_2 c_3 & s_2 s_3 & c_2 \end{bmatrix}$	$\theta_1 = t^{-1}(-C_{32} / C_{33})$ $\theta_2 = s^{-1}(C_{31})$ $\theta_3 = t^{-1}(-C_{21} / C_{11})$	$\pm\pi/2$
1-3-1	$\begin{bmatrix} c_2 & 0 & 1 \\ -s_2 c_3 & s_3 & 0 \\ s_2 s_3 & c_3 & 0 \end{bmatrix}$	$\frac{1}{s_2} \begin{bmatrix} 0 & -c_3 & s_3 \\ 0 & s_2 s_3 & s_2 c_3 \\ s_2 & c_2 c_3 & -c_2 s_3 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{13} / C_{12})$ $\theta_2 = c^{-1}(C_{11})$ $\theta_3 = t^{-1}(C_{31} / -C_{21})$	$0, \pm\pi$
1-3-2	$\begin{bmatrix} c_2 c_3 & -s_3 & 0 \\ -s_2 & 0 & 1 \\ c_2 s_3 & c_3 & 0 \end{bmatrix}$	$\frac{1}{c_2} \begin{bmatrix} c_3 & 0 & s_3 \\ -c_2 s_3 & 0 & c_2 c_3 \\ s_2 c_3 & c_2 & s_2 s_3 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{23} / C_{22})$ $\theta_2 = s^{-1}(-C_{21})$ $\theta_3 = t^{-1}(C_{31} / C_{11})$	$\pm\pi/2$
2-1-2	$\begin{bmatrix} s_2 s_3 & c_3 & 0 \\ c_2 & 0 & 1 \\ -s_2 c_3 & s_3 & 0 \end{bmatrix}$	$\frac{1}{s_2} \begin{bmatrix} s_3 & 0 & s_3 \\ s_2 c_3 & 0 & s_2 s_3 \\ -c_2 s_3 & s_2 & c_2 c_3 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{21} / C_{23})$ $\theta_2 = c^{-1}(C_{22})$ $\theta_3 = t^{-1}(C_{12} / -C_{32})$	$0, \pm\pi$
2-1-3	$\begin{bmatrix} c_2 s_3 & c_3 & 0 \\ c_2 c_3 & -s_3 & 0 \\ -s_2 & 0 & 1 \end{bmatrix}$	$\frac{1}{c_2} \begin{bmatrix} s_3 & c_3 & 0 \\ c_2 c_3 & -c_2 s_3 & 0 \\ s_2 s_3 & s_2 c_3 & c_2 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{31} / C_{33})$ $\theta_2 = s^{-1}(-C_{32})$ $\theta_3 = t^{-1}(C_{12} / C_{22})$	$\pm\pi/2$

α - β - γ Rotation Sequence	R	R ⁻¹	Angles as Function of Direction Cosines	Singular at $\theta_2 =$
2-3-1	$\begin{bmatrix} s_2 & 0 & 1 \\ c_2 c_3 & s_3 & 0 \\ -c_2 s_3 & c_3 & 0 \end{bmatrix}$	$\frac{1}{c_2} \begin{bmatrix} 0 & c_3 & -s_3 \\ 0 & c_2 s_3 & c_2 c_3 \\ c_2 & -s_2 c_3 & s_2 s_3 \end{bmatrix}$	$\theta_1 = t^{-1}(-C_{13} / C_{11})$ $\theta_2 = s^{-1}(C_{12})$ $\theta_3 = t^{-1}(-C_{32} / C_{22})$	$\pm\pi/2$
2-3-2	$\begin{bmatrix} s_2 c_3 & -s_3 & 0 \\ c_2 & 0 & 1 \\ s_2 s_3 & c_3 & 0 \end{bmatrix}$	$\frac{1}{s_2} \begin{bmatrix} c_3 & 0 & s_3 \\ -s_2 s_3 & 0 & s_2 c_3 \\ -c_2 c_3 & s_2 & -c_2 s_3 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{23} / -C_{21})$ $\theta_2 = c^{-1}(C_{22})$ $\theta_3 = t^{-1}(C_{32} / C_{12})$	$0, \pm\pi$
3-1-2	$\begin{bmatrix} -c_2 s_3 & c_3 & 0 \\ s_2 & 0 & 1 \\ c_2 c_3 & s_3 & 0 \end{bmatrix}$	$\frac{1}{c_2} \begin{bmatrix} -s_3 & 0 & c_3 \\ c_2 c_3 & 0 & c_2 s_3 \\ s_2 s_3 & c_2 & -s_2 c_3 \end{bmatrix}$	$\theta_1 = t^{-1}(-C_{21} / C_{22})$ $\theta_2 = s^{-1}(C_{23})$ $\theta_3 = t^{-1}(-C_{13} / C_{33})$	$\pm\pi/2$
3-1-3	$\begin{bmatrix} s_2 s_3 & c_3 & 0 \\ s_2 c_3 & -s_3 & 0 \\ c_2 & 0 & 1 \end{bmatrix}$	$\frac{1}{s_2} \begin{bmatrix} s_3 & c_3 & 0 \\ c_3 s_2 & -s_3 s_2 & 0 \\ -s_3 c_2 & -c_3 c_2 & s_2 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{31} / -C_{32})$ $\theta_2 = c^{-1}(C_{33})$ $\theta_3 = t^{-1}(C_{13} / C_{23})$	$0, \pm\pi$
3-2-1	$\begin{bmatrix} -s_2 & 0 & 1 \\ c_2 s_3 & c_3 & 0 \\ c_2 c_3 & -s_3 & 0 \end{bmatrix}$	$\frac{1}{c_2} \begin{bmatrix} 0 & s_3 & c_3 \\ 0 & c_2 c_3 & -c_2 s_3 \\ c_2 & s_2 s_3 & s_2 c_3 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{12} / C_{11})$ $\theta_2 = s^{-1}(-C_{13})$ $\theta_3 = t^{-1}(C_{23} / C_{33})$	$\pm\pi/2$
3-2-3	$\begin{bmatrix} -s_2 c_3 & s_3 & 0 \\ s_2 s_3 & c_3 & 0 \\ c_2 & 0 & 1 \end{bmatrix}$	$\frac{1}{s_2} \begin{bmatrix} -c_3 & s_3 & 0 \\ s_2 s_3 & s_2 c_3 & 0 \\ c_2 c_3 & -c_2 s_3 & s_2 \end{bmatrix}$	$\theta_1 = t^{-1}(C_{32} / C_{31})$ $\theta_2 = c^{-1}(C_{33})$ $\theta_3 = t^{-1}(C_{23} / -C_{13})$	$0, \pm\pi$

Roll-Pitch-Yaw Euler Angles

The most commonly used alternate set is the orientation angle used in aircraft dynamics: roll, pitch, yaw in order to remove the singularity ($\theta=0$).



Starting from the inertial frame the standard order is first yaw about the vertical axis (ψ), the pitch about the new wing axis (θ), and finally roll about the new longitudinal axis (ϕ). The complex – rotation matrix from the inertial to the body frame is then,

$$\begin{aligned}
 R^{ib} &= R_1(\phi)R_2(\theta)R_3(\psi) \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos \psi \cos \theta & \sin \psi \cos \theta & -\sin \theta \\ -\sin \psi \cos \phi + \cos \psi \sin \theta \sin \phi & \cos \psi \cos \phi + \sin \psi \sin \theta \sin \phi & \cos \theta \sin \phi \\ \sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi & -\cos \psi \sin \phi + \sin \psi \sin \theta \cos \phi & \cos \theta \cos \phi \end{bmatrix}
 \end{aligned}$$

We need the expression for the angular velocity written in the body-unit vectors.

$$\bar{\omega}^{\text{bi}} = \dot{\psi} \hat{k} + \dot{\theta} \hat{j}' + \dot{\phi} \hat{b}_1$$

$$\hat{k} = -\sin \theta \hat{b}_1 + \cos \theta \hat{k}''$$

$$\hat{k}'' = \cos \phi \hat{b}_3 + \sin \phi \hat{b}_2$$

$$= -\sin \theta \hat{b}_1 + \sin \phi \cos \theta \hat{b}_2 + \cos \phi \cos \theta \hat{b}_3$$

$$\hat{j}' = \cos \phi \hat{b}_2 - \sin \phi \hat{b}_3$$

$$\bar{\omega}^{\text{bi}} = \dot{\psi}(-\sin \theta \hat{b}_1 + \sin \phi \cos \theta \hat{b}_2 + \cos \phi \cos \theta \hat{b}_3) + \dot{\theta}(\cos \phi \hat{b}_2 - \sin \phi \hat{b}_3) + \dot{\phi} \hat{b}_1$$

$$= (-\dot{\psi} \sin \theta + \dot{\phi}) \hat{b}_1 + (\dot{\psi} \sin \phi \cos \theta + \dot{\theta} \cos \phi) \hat{b}_2 + (\dot{\psi} \cos \phi \cos \theta - \dot{\theta} \sin \phi) \hat{b}_3$$

$$\omega_1 = \dot{\phi} - \dot{\psi} \sin \theta$$

$$\omega_2 = \dot{\theta} \cos \phi + \dot{\psi} \sin \phi \cos \theta$$

$$\omega_3 = -\dot{\theta} \sin \phi + \dot{\psi} \cos \phi \cos \theta$$

$$\therefore \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

The roll (ϕ), pitch(θ) and yaw(ψ) system becomes singular for a pitch angle of $\theta=90^\circ$ (vertical flight: uncommon for a long period of time).

The choice of an appropriate system can thus depend on the particular orientations of interest.